

Well-foundedness of an Original Infinite Basic Laver Pattern under I3

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Abstract

We prove, in ZFC assuming I3, that the original infinite basic Laver pattern tree rooted at

$$R = (1, 0)1(2, 1, 0)1(3, 2, 1, 0)2(4, 3, 2)1(5, 4, 3, 2)2(6, 5, 4)1$$

is well founded. We use the original full-tail copy, mark filter, frozen mark scan, and staircase native completion. Successor nodes have their unique cut child. The proof does not calculate the ordinal represented by the root. A finite certificate assigns inaccessible point ordinals and bounded elementary rank maps to the rows. We prove closure of these certificates, including the marked packet rules, and extend each terminal bounded map to an internal ultrapower. A completed child has cap strictly below the image of its parent's cap. Since the certificate predicate is first order, a least bad cap then gives a contradiction by elementarity. No infinite iteration strategy, temporal direct-limit assumption, or unproved local closure theorem is used.

1. Statement and scope

An IBLP expression is finite. The adjective *infinite* refers to its unbounded family of finite fundamental-sequence expansions. We consider the descendant tree whose edges are the operations specified below, starting from R . A branch is a sequence of successive descendants; well-foundedness means that no infinite branch exists.

Theorem 1. *ZFC + I3 proves that this descendant tree is well founded.*

Here I3 asserts the existence of a nontrivial elementary embedding

$$g : V_\lambda \longrightarrow V_\lambda, \quad \lambda = \sup_{i < \omega} g^i(\text{crit}(g)).$$

The specification in [1] assigns positive-parameter expansions only to nonsuccessors. Accordingly, a successor has just its cut child; equivalently, all its parameter labels may denote that same child. This convention does not alter any fullcopy or completion rule. Zero has no children. We do not replace the original staircase by an all-points packet or replace accurate traces by a branching rule.

The proof has two distinct parts. A finite event induction establishes the exact combinatorial and semantic closure needed by the algorithm. The global argument uses one finite expansion from a least bad certificate. In particular, it never assumes that an arbitrary sequence of rank-to-rank embeddings has a well-founded direct limit.

2. The precise recursive tree

2.1 Rows, traces, and ordinary shapes

We write rows increasingly, using zero-based positions. A pattern with n rows consists of finite increasing integer sequences A_r , positive steps L_r , and sets $M_r \subseteq A_r$, where

$$\min A_r \geq 0, \quad |A_r| \geq 2, \quad \max A_r = r.$$

The displayed descending notation in the theorem reverses these rows; the integer after a parenthesis is its step. Put

$$p(r) = A_r[-L_r - 1], \quad e(r) = A_r[-L_r], \quad q(r) = A_r[-2].$$

If a marked point $b = A_r[j]$ has $j \geq L_r$, its paired source is $\delta = A_r[j - L_r]$. Follow p from b . If the resulting strictly decreasing chain reaches δ , its **accurate trace** is

$$b = v_0 > v_1 > \dots > v_{d-1} = t > v_d = \delta, \quad p(v_i) = v_{i+1}.$$

Otherwise the trace is undefined. The last nonterminal vertex t is its bottom factor.

The following row shapes and mark window will be preserved from R :

Shape	Length and step
Long	3, with step 1
Medium	$2L$, with $L \geq 1$
Short	$2L - 1$, with $L \geq 3$

Every mark has zero-based position

$$L + 1 \leq j \leq |A_r| - 2. \quad (2.1)$$

These are invariants of this root's generated domain, rather than additional tests inserted into the expansion algorithm. In particular, long rows and two-column rows have no marks.

2.2 Cut and fullcopy

Cut deletes the last row. A nonzero pattern is a successor precisely when its last row has two columns and minimum 0. For a nonsuccessor with last row $C = A_n$, minimum a , step L , and $p = p(n)$, define

$$f(x) = \begin{cases} x, & x < a, \\ C[j + L], & x = C[j] < p, \\ x + n - p, & x \geq p, \end{cases} \quad (2.2)$$

and leave all other values undefined. Fullcopy retains rows $1, \dots, n - 1$ and appends the images of rows $p, \dots, n$, including the old last row. Thus its new length is $2n - p$. All required column images must exist; otherwise the first copy is undefined. Steps are unchanged. For the ordinary nonsuccessors under consideration, $p \geq 1$.

For a source mark y , first require its old accurate trace. Keep its image $f(y)$ exactly in one of these cases:

1. Its bottom factor is at least p .
2. Its first trace vertex $u < p$ satisfies $u < a$.
3. That first vertex is a selected controller column, $f(u) \in M_n$, and the following position test holds.
If $f(y)$ has zero-based position j in the copied row of step L_r , then

$$A'_{f(r)}[j + 1 - L_r] \leq a. \quad (2.3)$$

For a mark with an accurate trace the indicated position exists. This is precisely the one-based test of [1] translated to zero-based positions. No additional marks are retained.

2.3 A macro expansion and its frozen scan

For $m = 0$, set $E(A, 0) = \text{cut}(A)$. For a nonsuccessor and $m \geq 1$, if the first fullcopy is undefined, also use cut . Otherwise perform m fullcopies, cut the final self-row, and scan from the original row index n . We prove below that every later fullcopy is then defined.

The temporary map Rec starts empty. At a current old row r , freeze its current marked candidates and process them in increasing order. For a candidate b , skip it unless its trace is accurate and its bottom factor t has a nonempty record. For every internal trace edge $p(u) = v$, with v still nonterminal, additionally require

$$v + 1 \in A_u. \quad (2.4)$$

If these tests pass, let $S = \text{Rec}(t)$, $h = |S|$. Insert S and $b + 1, \dots, b + h$ into the row, sort and deduplicate, increase its step by h , unmark S , and mark the latter interval. Other marks remain. Newly produced marks are not reconsidered during this frozen pass.

Next perform native completion. Long rows do nothing. Otherwise start at $e = e(r)$, repeatedly replace the current point u by $q(u)$ while $q(u) > p(r)$, and record the visited replacement points. Let their increasing enumeration be

$$p(r) < s_1 < \dots < s_h < e(r).$$

If $h = 0$, do nothing. Otherwise store this record at the old base r , insert h rows immediately after it, and shift all later indices and columns accordingly. Form the top family row by adjoining S and $r + 1, \dots, r + h$ to the old row and increasing the step by h . Keep old marks, mark $r, \dots, r + h - 1$, and unmark S and the last point.

Recover the lower family rows backwards. From row $r + i$, delete its last explicit point $r + i$ and its current e -column, decrease the step by one, and remove the mark on $r + i - 1$. Both deletion positions are selected before deletion. There is exactly one exception: when the original row was medium and $i = h$, retain the e -column and the step, deleting only the last explicit point and removing the specified mark. Advance the scanner to $r + h + 1$. Hence new family rows are skipped. At the end discard the records.

This is a recursive finite operation: the scan processes only its finitely many entrance-old rows, and each individual record is a strictly decreasing finite chain.

3. Finite bounded certificates

Work temporarily in a transitive model $M \models \text{ZFC}$, with all ranks interpreted internally. A certificate consists of inaccessible ordinals

$$\theta_0 < \dots < \theta_n < \theta_{n+1} = \Theta$$

and set maps

$$K_r : V_{\theta_{e(r)}+1}^M \longrightarrow V_{\theta_{r+1}+1}^M. \quad (3.1)$$

Each map is elementary, has critical point $\theta_{\min A_r}$, and realizes all step edges, including the implicit endpoint. Explicitly, append $a_\ell = r + 1$ to $A_r = (a_0, \dots, a_{\ell-1})$; require

$$K_r(\theta_{a_i}) = \theta_{a_{i+L_r}} \quad (0 \leq i \leq \ell - L_r). \quad (3.2)$$

In particular $K_r(\theta_{p(r)}) = \theta_r$ and $K_r(\theta_{e(r)}) = \theta_{r+1}$. Require the shapes and mark window in Section 2, accurate traces for every mark, and numerical saturation

$$q(e(r)) \leq p(r) \quad \text{for every row } r. \quad (3.3)$$

We next specify the weak-mark clause using these bounded graphs alone. For $K : V_{\alpha+1} \rightarrow V_{\beta+1}$, put

$$K^\#(z) = K(z \cap V_\alpha), \quad \rho_K(\eta) = K(\min(\alpha, \eta)). \quad (3.4)$$

Both expressions are defined without choosing a full extension. For a finite word $W = K_1^\# \circ \dots \circ K_d^\#$, its rank map is the corresponding composite of the ρ 's. Its natural cap is obtained recursively from the innermost target, applying those saturated rank maps outwards.

For an accurate trace, consecutive factors satisfy $\theta_{v+1} \leq \theta_{e(u)}$ whenever $p(u) = v$. Form its actual bounded composition by restricting each outer factor to the rank structure which is the preceding composite's image. Relativizing elementarity to that source rank shows that the restriction is elementary onto the corresponding image rank; no elementary inclusion between two different rank structures is assumed. Denote the resulting composite by W . Its source endpoint is $\alpha_t = \theta_{e(t)}$, and its projection operator is the composite of the factors' $\#$ operators. Its natural cap is

$$\Delta = \rho_W(\alpha_t) = W(\alpha_t), \quad \theta_b < \Delta \leq \theta_{b+1}. \quad (3.5)$$

The mark clause is

$$K_r^\#(z) \cap V_\Delta = W^\#(z) \quad \text{for all } z \in V_\Delta. \quad (3.6)$$

Write $F \equiv_\delta G$ when their output intersections with V_δ agree on all inputs from V_δ .

Lemma 2 (bounded weak calculus). For every bounded word,

$$W^\#(z \cap V_\eta) = W^\#(z) \cap V_{\rho_W(\eta)}. \quad (3.7)$$

Cutting its factors down to smaller source ranks replaces its output by its intersection with the new natural cap. At a limit cutoff δ , weak equality tested only on $x, z \in V_\delta$ is equivalent to equality of output intersections with V_δ for **all** inputs $z \in M$. Right composition preserves such equality; left composition by H transports its cutoff to $\rho_H(\delta)$.

Proof. For one factor, (3.7) is elementarity applied to intersection with $V_{\min(\alpha, \eta)}$. In particular a restriction at $\alpha' \leq \alpha$ has operator $K^\#(z) \cap V_{K(\alpha')}$. Induct through the word; simultaneous restrictions introduce the minimum of the transported restriction caps. Each rank map is nondecreasing and $\rho_W(\eta) \geq \min(\eta, D_W)$, where D_W is the natural cap. For $x \in V_\delta$, choose $\text{rank}(x) < \eta$ with $\eta + 1 < \delta$. Replacing an arbitrary input z by $z \cap V_\eta \in V_\delta$ does not change membership of x in an output, by (3.7); if x is above its cap it is absent anyway. This proves the stronger input formulation. The composition assertions now follow by applying (3.7) to equal truncated sets. ■

4. Extending one bounded row internally

The following elementary extender construction is what permits the model to change without requiring a full owner graph in the new model.

Lemma 3. Suppose M is a transitive proper-class inner model of ZFC, externally closed under countable sequences, and $K \in M$ is an elementary map $V_{\alpha+1}^M \rightarrow V_{\beta+1}^M$, where α, β are inaccessible in M , $K(\alpha) = \beta$, and its critical point exceeds ω . There is an internal extender ultrapower

$$I : M \longrightarrow N$$

which is externally well founded and countably closed, and satisfies

$$V_{\beta}^N = V_{\beta}^M, \quad I \restriction V_{\alpha+1}^M = K. \quad (4.1)$$

Proof. Let $X = V_{\alpha}^M$. For finite tuples a of seeds in V_{β}^M , derive

$$E_a = \{B \subseteq X^{|a|} : B \in M \text{ and } a \in K(B)\}.$$

Use the usual finite-product coding inside X . Every test relation is coded by a subset of X , so belongs to the domain of K . The coherent seed system is a set in M , and forms the internal ultrapower using all M -functions on its source domains.

For external well-foundedness, a purported countable descent has representatives $[f_i, a_i]$. Countable closure puts the sequences of functions and seeds in M . Regularity of β there allows the seeds to be combined into one seed $a_* \in V_{\beta}^M$; fixed source projection functions recover its coordinates. Rewrite every adjacent membership comparison over this common seed, obtaining sets B_i . The sequence of top-rank subsets need not itself belong to $V_{\alpha+1}$. Instead encode it by the single relation

$$C = \{(i, x) : i < \omega, x \in B_i\} \subseteq V_{\alpha}^M.$$

Apply K to this relation. Since K fixes ω , its slices are the $K(B_i)$, and it preserves their countable intersection as defined from C . The master seed belongs to every slice, so the source intersection is nonempty. A point in it yields an actual infinite membership descent in M , contradicting Foundation.

After collapse, the elements of $I(X)$ have representatives with values in X . Their function graphs are subsets of X , so

$$[f, a] \mapsto K(f)(a)$$

defines an $\in$ -isomorphism onto V_{β}^M . Derived relations prove well-definedness and injectivity, and identity representatives give every seed. Both sets are transitive, so the isomorphism is the identity. Hence $I(\alpha) = \beta$ and $I(X) = V_{\beta}^M$. For every $x \subseteq X$, seed membership gives $a \in I(x) \iff a \in K(x)$ for all $a \in V_{\beta}^M$. Thus agreement extends to the entire $V_{\alpha+1}^M$, including its top subsets.

The same argument, applied to its own countable sequences, proves well-foundedness internally in M . The internal collapse is therefore computed in M and agrees with the external collapse, so $N \subseteq M$. For countable closure of N , combine representatives $[f_i, a_i]$ over the master seed and use the one function $x \mapsto \langle f_i(\pi_i(x)) : i < \omega \rangle$ in M . Its ultrapower value is the given countable sequence.

Finally, choose in M a bijection $h : \alpha \rightarrow V_{\alpha}^M$. Its coded graph is in $V_{\alpha+1}^M$, and $I(h) = K(h) : \beta \rightarrow V_{\beta}^M$. Every rank seed is generated by an ordinal seed below β . The corresponding ordinary ordinal-seed extender has this same ultrapower and belongs to M . ■

Notice that (4.1) does not claim equality of $V_{\beta+1}^N$ and $V_{\beta+1}^M$. That stronger assertion is unnecessary.

5. Raw fullcopy, including its mark filter

Lemma 4. If the first fullcopy is defined, every prescribed finite number of successive fullcopies is defined.

Proof. Its domain is $[0, a) \cup (C \cap [a, p)) \cup [p, \infty)$, on which f is strictly increasing and maps its domain into itself. Indeed, if $a < p$, ordinary length $\ell \leq 2L+1$ gives $f(a) = C[\ell] \geq C[\ell-L-1] = p$; if $a = p$, then $f(a) = n \geq p$. Thus all nonfixed images lie in the translation interval. Write $d = n - p$. The new last row is $f(C)$, its new p is n , and its next copying map f' satisfies

$$f'(f(x)) = f(f(x))$$

on the entire old domain: below a both sides are fixed; on selected columns the same positional step applies; above p both translate by d . The next source rows are images of already legal rows. Induction proves the assertion. ■

Apply Lemma 3 to the old last graph, with $\alpha = \theta_{e(n)}$, $\beta = \Theta$. Retain prefix graphs in N . Their targets are at most $\theta_n < \Theta$, so their graphs have rank below Θ , and their rank structures are unchanged. Give each copied row the graph $I(K_r)$. For a copied column $x \geq p$ use the point $I(\theta_x)$; selected columns below p satisfy the same equation by agreement with the last graph, and columns below a are fixed. The glue is $I(\theta_p) = \theta_n$. This constructs the entire new point table and every step edge in N .

We verify weak marks rather than assuming a full owner exists. Write IF for the transported bounded word. For every $z \in M$,

$$(IF)^\#(I(z)) = I(F^\#(z)), \quad I(z) \cap V_\Theta = K_n^\#(z). \quad (5.1)$$

These follow from elementarity and (4.1). If $\kappa = \theta_a = \text{crit}(I)$, then $I(z) \cap V_\kappa = z \cap V_\kappa$. Consequently, for a bounded word H in N and $z \in N \subseteq M$,

$$H^\#(I(z)) \cap V_{\rho_H(\kappa)} = H^\#(z) \cap V_{\rho_H(\kappa)}. \quad (5.2)$$

For a retained mark there are exactly the three cases in Section 2. In the all-high case every factor is transported, so elementarity preserves its weak certificate. In the low case write the old trace as $U \circ V$, where the retained suffix begins below a . Its cap $D_V \leq \kappa$. Elementarity fixes its weak action below D_V , so $IV \equiv_{D_V} V$. Lemma 2 transports this equality through IU . The new cap is no larger than the transferred old cap, since $I(D_V) \geq D_V$. Equality here is not assumed when $D_V = \kappa$.

For the mixed case, let $u \in [a, p)$ be the first lower trace vertex, $v = f(u) \in M_n$, and let T be the controller's accurate mark trace from v to u , with cap τ . The old row $F = K_r$ satisfies $F \equiv_\delta U \circ V$, and $K_n \equiv_\tau T$. The next-source test (2.3), together with strict increase of f , makes the old paired source d_0 satisfy $d_0 < a$ and $f(d_0) = d_0$. The new trace is therefore exactly

$$(IU) \circ T \circ V : \quad (5.3)$$

the copied segment reaches v , the old controller trace reaches u , and the retained suffix reaches d_0 . Equations (5.1), (5.2), and Lemma 2 identify the new row with (5.3) below

$$\min\{\rho_{IF}(\kappa), I(\delta), \rho_{IU}(\tau)\}. \quad (5.4)$$

For completeness, start with $(IF)^\#(z)$, insert I on its input using (5.2), rewrite as $I(F^\#(z))$, use the old weak equality, move I past U by (5.1), and replace its action on the remaining argument below τ by $T^\#$.

All three bounds cover the new natural cap D . Indeed weak equality on the ordinal D_V gives

$$\rho_T(D_V) = \min(\rho_{K_n}(D_V), \tau) \leq I(D_V).$$

Thus $D = \rho_{IU}(\rho_T(D_V)) \leq I(\delta)$, and also $D \leq \rho_{IU}(\tau)$. Finally (2.3) places the next source column at or below κ . The copied row sends that column to the next target after the mark, which is at least the mark's immediate upper point and hence at least D . Therefore $\rho_{IF}(\kappa) \geq D$. This proves the mixed weak clause using the exact original filter.

6. Why every actual mark packet fits

We now prove the needed numerical capacity theorem. No auxiliary tree order or comparison between bridge-mark sources is required.

Fix a saturated parent of length n , let $p = p(n)$, $d = n - p$, and perform $m \geq 1$ fullcopies followed by cut. Keep fixed labels for the old vertices present at scan entrance. The scanned rows form blocks

$$B_h = [n + (h - 1)d, n + hd - 1], \quad 1 \leq h \leq m. \quad (6.1)$$

The h -th block consists of the h -th copies of original rows $p, \dots, n - 1$. Fresh native-family vertices are not entrance-old vertices.

The subscript 0 always refers to these fixed entrance labels. Expressions such as $v + j$ below use the current numerical position of that old vertex and its fresh family, after all intervening shifts.

Lemma 5 (active origins). Every frozen mark that can use a nonempty record has all its nonterminal trace factors in its own block. Moreover, if $u, p_0(u) \in B_h$, then $e_0(u) \in B_h$ and

$$q_0(e_0(u)) \leq p_0(u). \quad (6.2)$$

Proof. A mark whose original bottom factor is at least p always uses the all-high copy alternative. Its entire nonterminal trace is translated into its block. A mark with bottom below p , if retained by either other alternative, keeps that bottom: the mixed splice of Section 5 rejoins the same old suffix. Every later copy also retains this suffix. Indeed the next controller minimum is at least the preceding p , since the ordinary row length is at most $2L + 1$. Such a bottom remains in the unscanned prefix and receives no record. This proves the first assertion. For the second, $p_0(u)$ in its block means that its original p -column was in the copied interval. Its next column e and the row at that column were therefore copied by the same map. Parent saturation transfers to (6.2). Long and two-column step-1 rows satisfy it identically. ■

Partition each block by descending entrance p_0 -chains. Write $\sigma(u)$ for the last vertex of that chain still in its block; vertices with the same σ form a component. The term denotes only this finite functional relation.

We use induction in the actual scan order. At a current event, all strictly earlier marked events have already been proved to insert fresh paired packets, preserve old p/e up to native, preserve old paired sources, and place their right packets inside their target's same-scan family. Native independently preserves p and q of its old lowest row; its new vertices satisfy $q(v + j) = v + j - 1$.

Lemma 6 (inheritance and component width). If native at $u \in B_h$ has a nonempty record and $v = p_0(u) \in B_h$, there was an earlier successful frozen mark v at the old row $c = e_0(u)$. If that mark used the bottom record at z , then

$$\text{Rec}(u) \text{ increasing} = [v + 1, \dots, v + H], \quad H = |\text{Rec}(z)|, \quad (6.3)$$

where z lies on v 's chain in the same block. Consequently every nonempty record at u has the width of a nonempty record at $\sigma(u)$; in particular that seed record exists.

Proof. Nonempty native excludes the long and two-column cases, so $c < u$ is already processed. By (6.2), $q_0(c) \leq v$. If the inequality is strict, every old proper target of c lies before the old vertex

v . Earlier correct packets remain in their own earlier families, and native preserves the old base's q ; hence $q(c) < v$ still holds, making native at u empty. Thus $q_0(c) = v$. The only event at c capable of making $q(c) > v$ is its frozen mark v , the largest old proper target. If it uses H points, it sets $q(c) = v + H$. Native at c preserves that value. The consecutive family chain at v now gives exactly (6.3). Lemma 5 puts its bottom z in this block, and accuracy puts z on v 's p_0 -chain.

Induct on processed old vertices. If $p_0(u)$ is outside its block, then $u = \sigma(u)$. Otherwise (6.3) reduces its width to the earlier vertex z , which has the same component seed. This proves the final assertion. Only strictly earlier successful mark events were used. ■

Corollary 7 (exact capacity). Suppose the current mark has bottom record of width $h > 0$ and passes all guards (2.4). Every nonterminal factor has a nonempty record of exactly width h . For each internal edge $p(u) = v$, its increasing record is $[v + 1, \dots, v + h]$. Thus the head packet lies in its own same-scan native family, and, writing the bottom record increasingly as $s_1, \dots, s_h$, the new traces are

$$b + j \longrightarrow \dots \longrightarrow t + j \longrightarrow s_j, \quad 1 \leq j \leq h. \quad (6.4)$$

Proof. Start at t and move upwards. If v has a nonempty record, $v + 1$ is fresh. The guard places it in the processed row u with $p(u) = v$, so its current e -column is $v + 1$. Correct previous marks preserved its old e until native; hence native at u was nonempty. Lemmas 5 and 6 put these records in one component with equal width. Equation (6.3) gives their exact prefixes, and the native family formula gives (6.4). ■

7. Completion closes the bounded certificates

7.1 Native restrictions

Let the native row's graph be $K : V_{\alpha+1} \rightarrow V_{\beta+1}$, and its increasing packet be $s_1, \dots, s_h$. Insert the point ordinals

$$\xi_j = K(\theta_{s_j}), \quad \theta_r < \xi_1 < \dots < \xi_h < \beta.$$

These are inaccessible: the bounded source and target rank structures are correct about inaccessibility of their internal cardinals, and K is elementary. The graph on family row $r + j$ is exactly

$$K \upharpoonright V_{\alpha_j+1}, \quad \alpha_j = \theta_{s_{j+1}} \ (j < h), \quad \alpha_h = \alpha. \quad (7.1)$$

Its target is its required next point. The top row's old and new step edges follow from the paired insertion. In a short row the deleted central column lies after all retained sources and before all targets, so removing it decreases the step by one. In the medium exception that column is already the first target and must remain. Deleting the last explicit point makes it the lower row's implicit endpoint. These observations prove (7.1) for every reversed family row.

Put $s_0 = p$. Row $r + j$ marks $r + k$, for $k < j$, directly from s_k . The factor at $r + k$ is a restriction of the carrier at $r + j$; Lemma 2 proves its weak equality at its natural target cap. Old marks keep their sources and their old p -traces, while their carrier and previously processed factors are restrictions of their scan-entry graphs. Their weak clauses follow by the same cutback lemma.

The base preserves its old p, q , and has new $e = s_1$, with $q(s_1) \leq p$. Higher family rows have $p = s_j, e = s_{j+1}$, or the old e at the top, with the corresponding $q(e) = p$. Thus every family row is saturated. The medium exception produces a short step at least 3; other shapes remain ordinary. A nonempty medium native has old step at least 2. Its earliest new marked source has positive position $L - 1$; for a short row it has position $L - 2 \geq 1$. This proves (2.1) for the native marks.

7.2 Marked packets and their fresh gaps

Freeze the scan-entry graphs as finite proof data. For a current old mark b , let W be its historical trace word, ending in t , with cap $\Delta = W(\alpha_t)$. Earlier correct marks preserve this row's central endpoints, so its graph K_r is still unchanged. Let $\theta_{\delta+}$ be the next source column after its paired source θ_δ . Ascending frozen-mark processing leaves its next target at least the historical $\theta_{b+1} \geq \Delta$. Thus

$$K_r(\theta_{\delta+}) \geq \Delta.$$

If $\theta_{\delta+} < \alpha_t$, then $W(\theta_{\delta+}) < \Delta$. Use this latter ordinal as a membership test in the historical weak equality: it belongs to the former ordinal image but not to itself. This contradiction proves

$$\alpha_t \leq \theta_{\delta+}. \quad (7.2)$$

The bottom native record therefore lies strictly in the carrier's source gap. Corollary 7 and the native definitions give actual named point identities

$$W(\theta_{s_j}) = \theta_{b+j} < \Delta.$$

Weak equality on these ordinal inputs then gives

$$K_r(\theta_{s_j}) = \theta_{b+j}. \quad (7.3)$$

The image packet lies strictly between θ_b and its next old target. Thus both packets are fresh, disjoint, and in their correct paired gaps; deduplication removes nothing.

For clarity, the array alignment also requires checking. If the old mark position is q , insert h sources after $u = q - L$ and h targets after q , and use step $L + h$. Sources at or before u are unshifted and their old targets shift h ; later sources shift h and their targets shift $2h$. All old edges therefore remain, and (7.3) supplies the new ones. Window (2.1) puts the source gap before the central p/e and the target gap after e . Those endpoints, the graph K_r , and every old mark's paired source are unchanged. New paired sources are above a positive old source position, so the window persists.

Finally the factors on (6.4) are precisely restrictions of the historical factors of W . By Lemma 2 their word is $W^\#(z) \cap V_{\Delta_j}$, where $\Delta_j \leq \Delta$ is its new natural cap. The historical row equality therefore proves each new mark's weak clause. The carrier cap exceeds $\theta_{b+j+1} \geq \Delta_j$. Old marks obtain any required smaller cutoff by restriction. Subsequent native is already covered above.

This closes the event induction: capacity uses correctness only of earlier events; the current event obtains its gap correctness and weak clauses afterwards. Each processed row becomes saturated. Later events cannot change its earlier central rows, and the unscanned prefix was already saturated. Hence the whole completed child has a certificate with the same cap as the scan entrance.

8. The least bad cap and the initial root

Let $\text{Good}_R(P, \Theta)$ mean that P is finitely reachable from R and has the certificate of Section 3 with cap Θ . This is a single first-order predicate of set theory. Elementary maps between set rank structures are expressed using their set satisfaction relations. Every other clause concerns finitely many sets, bounded weak equalities, or an arithmetical computation. Neither external countable closure nor a proper-class embedding is included in this predicate.

One legal positive macro, performed through the internal ultrapowers of Lemma 3, has finite composite $I : M \rightarrow M'$. Every intermediate model remains externally well founded and countably closed. After the last copy, cutting its final self-row changes the cap to

$$\Theta' = I(\theta_n) < I(\Theta). \quad (8.1)$$

Completion does not change this cap. Retained graphs were carried unchanged in the preserved lower rank; copied graphs are internal images; native uses their restrictions. Thus the child's final graph is already in M' . It is not reintroduced by assuming a fresh full realization. A cut alone gives cap $\theta_n < \Theta$.

To finish, suppose there is a bad certified descendant, where bad means having an infinite branch. Choose the least cap Θ of such a descendant in V , and fix its certificate and a branch real. A cut child immediately contradicts minimality. Otherwise apply the one finite macro above, starting with $M = V$. Elementarity says that M' regards $I(\Theta)$ as the least cap of a bad Good_R -certificate. All embeddings fix every real, so the same branch tail belongs to M' and still witnesses badness of the child; its verification is arithmetical and absolute. The child certificate in M' has the smaller cap (8.1), contradiction.

It remains to supply the root certificate. Let g witness I3, write $\kappa_i = g^i(\text{crit}(g))$, and set $\theta_i = \kappa_i$ for $0 \leq i \leq 7$. Let $*$ denote application, and let powers denote ordinary composition. Use the following owners, restricted to their required source ranks plus one:

Row	Increasing core	Step	Owner
1	$(0, 1)$	1	g
2	$(0, 1, 2)$	1	g
3	$(0, 1, 2, 3)$	2	g^2
4	$(2, 3, 4)$	1	$g^2 * g$
5	$(2, 3, 4, 5)$	2	$g^2 * (g^2)$
6	$(4, 5, 6)$	1	$g^4 * g$

Application is the usual union of the images of bounded restrictions. Its elementary maps satisfy $(j * k) \circ j = j \circ k$ and distribute over composition in the right argument [2, Section 1]. Put $g_s = (g^+)^s(g)$, where $g^+(k) = g * k$. Then $\text{crit}(g_s) = \kappa_s$ and $g_s(\kappa_t) = \kappa_{t+1}$ for $t \geq s$, since $g_s \circ g^s = g^{s+1}$. The fourth and sixth owners are g_2, g_4 , and the fifth is g_2^2 . These identities verify every row edge, including the implicit endpoints $\kappa_2, \dots, \kappa_7$. All points are inaccessible, marks are empty, and $q(e(r)) = p(r)$. The root is therefore certified under I3. The contradiction above proves Theorem 1. ■

The same model-changing cap inequality can alternatively be organized using the well-founded pointed-extender order in Goldberg [3]. The direct least-bad-cap proof given here avoids that additional theorem. The conclusion concerns the specified descendant tree and does not calculate its ordinal height or assert a value for it.

References

1. Renrui Qi, *Infinite Basic Laver Patterns (IBLP): a formal specification*, supplied four-page manuscript, Definitions 1–15.
2. Randall Dougherty, *Critical points in an algebra of elementary embeddings, II*, particularly Section 1, [arXiv:math/9503204](https://arxiv.org/abs/math/9503204).
3. Gabriel Goldberg, *Rank-to-Rank Embeddings and Steel's Conjecture*, particularly the pointed-extender well-foundedness theorem, *author's manuscript*.